

GRB 081222

AGENTIC AI REPORT BY VIKAS CHAND, KHUSHBOO SHARMA, AND JAGDISH C. JOSHI

ABSTRACT

We present the complete timing and time-resolved spectral analysis of the single-pulse burst GRB 081222 (*Fermi*/GBM trigger bn081222204; $z = 2.77$), the second burst of a 106-burst single-pulse campaign run on a staged, gate-reviewed pipeline, and the campaign’s first burst with a redshift. The windowed duration is $T_{90} = 11.33 \pm 0.32$ s (3.01 s rest frame) with only 2.4σ of emission beyond the analysis window; per-band durations follow a poor single power law ($T_{90} \propto E^{-0.11 \pm 0.04}$, $\chi^2/\text{dof} = 4.3$). The pulse is FRED-like (two-sigmoid asymmetry $\varphi = 0.263 \pm 0.029$). The minimum variability timescale is estimator-dependent by construction: a windowed method finds 30.4 ± 2.1 ms (≈ 8 ms rest frame), a global wavelet method returns 215 ± 17 ms, and the Haar cross-check yields only an upper limit (< 1.02 s). Scanning the cross-correlation fit window around the peak gives a spectral lag of $\tau = 0.474_{-0.269}^{+0.171} \pm 0.377_{\text{win}}$ s (25–50 vs. 100–300 keV, soft lagging hard): this burst’s flat-topped correlation localizes its peak only to ± 0.4 s, and we quote that as a systematic. We then fit 24 spectral models in six Bayesian-block bins and the time-integrated spectrum. The low-energy index runs $\alpha = -0.70$ to -0.96 — synchrotron-compatible, in sharp contrast to the first campaign burst — and the Band function wins the integrated spectrum outright. In the decay phase a cut-off power law plus blackbody with a physical temperature ($kT = 16.5 \pm 3.7$ keV) tops the 24-model ranking by $\Delta\text{AIC} = 0.8$; it passes our Wien-peak edge gate but fails a residual-evidence test, and we report it as a tracked candidate, not a detection. Every panel of the model grid is rendered under a no-model-dropped rule with three provenance tiers; all figures and numbers passed independent verification gates.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

GRB 081222 triggered the *Fermi* Gamma-ray Burst Monitor (Meegan et al. 2009) on 2008 December 22 (RA = 22°7, Dec = −34°1), with a GBM-catalog duration of 18.9 ± 2.3 s and a 10–1000 keV fluence of 1.19×10^{-5} erg cm^{−2}. A secure redshift $z = 2.77$ (Cucchiara et al. 2008; Graham et al. 2008) makes it the first burst of this campaign with rest-frame leverage, and *Konus-Wind* provides an independent time-integrated spectrum (Golenetskii et al. 2008). As burst #2 of the single-pulse campaign, it also carries a methodological role: every step below runs on the pipeline frozen from burst #1 (see Band et al. 1993; Yu et al. 2019), and the two bursts already bracket opposite answers to two physical questions — the low-energy index’s relation to the synchrotron limit, and the presence of thermal components.

2. OBSERVATIONS, DETECTORS, AND APPROVED SELECTIONS

We use TTE data from NaI detectors n_0 (reference, engine-selected), n_1 , n_2 , and BGO b_0 (Figure 1), with the spectral convention of Chand et al. (2020): NaI 8–900 keV excluding the 30–40 keV K-edge, BGO 0.2–38 MeV. Background and source intervals are the human-approved Stage-1 selections (source window (−1.16, 16.48) s; Figures 2–3); the same selections drive spectroscopy and timing. Inter-detector effective-area factors float in [0.8, 1.2]; we note at the outset that b_0 ’s factor rails at 0.80 in most fits (and 1.20 in the last bin) — a calibration prior held fixed by design, and a caveat that travels with every low-energy statement in this paper.

3. TIMING ANALYSIS

All timing runs on the approved selections, reference NaI n_1 for the temporal chain.

3.1. Durations

The windowed duration is $T_{90} = 11.33 \pm 0.32$ s spanning [0.53, 11.86] s, with the covariance-true Monte

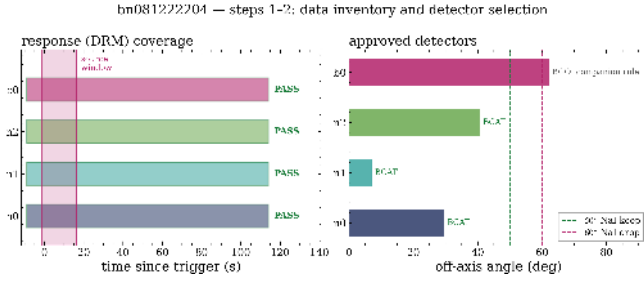


Figure 1. Detector selection for GRB 081222: source angles with the 50° rule; selected detectors n_0 , n_1 , n_2 , b_1 .

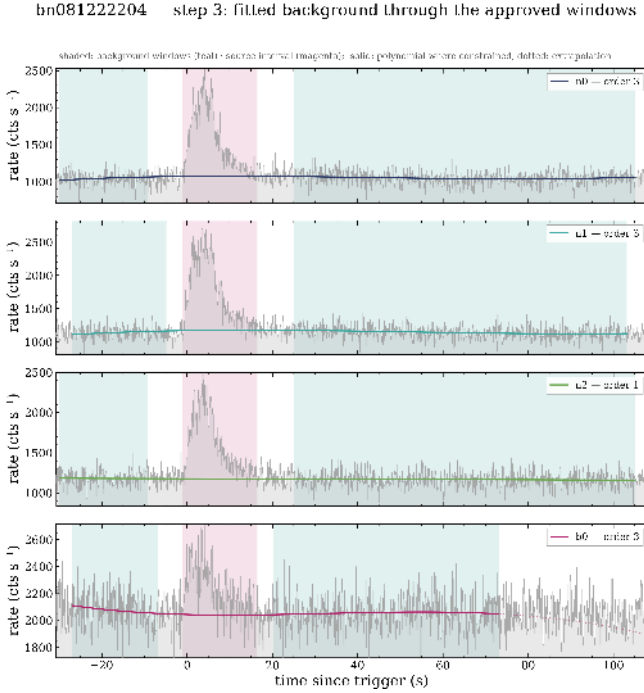


Figure 2. Approved background intervals and per-detector polynomial fits.

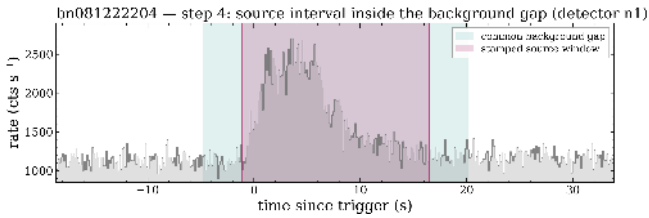


Figure 3. Approved source interval over the background-subtracted light curve.

Carlo of the campaign estimator; the interval beyond the source window holds only 2.4σ of excess, so unlike burst #1 this window captures the burst and the duration is not flagged as a lower limit. The GBM catalog quotes 18.9 ± 2.3 s: the difference is the catalog’s own (wider) interval and fluence-space method versus

our windowed count-space measurement, and we flag it for the literature-comparison step rather than reconcile it by hand. $T_{50} = 4.14$ s. In the rest frame, $T_{90}/(1+z) = 3.01$ s.

Per-band durations (Figure 4) run 11.5 ± 0.9 s (8–25 keV) to 8.1 ± 0.7 s (100–350 keV) but follow a single power law poorly ($T_{90} \propto E^{-0.11 \pm 0.04}$, $\chi^2/\text{dof} = 4.3$): three soft bands are nearly equal before a drop, a caution against per-burst use of the population-mean band scaling (Qin et al. 2013).

3.2. Pulse Shape

The Kocevski et al. (2003) profile wins the shape comparison ($\chi_r^2 = 1.09$; Norris 1.12), and the two-sigmoid of Gowri et al. (2025) gives $\varphi = s_l/s_r = 0.263 \pm 0.029$ ($R^2 = 0.937$): FRED-like, and statistically indistinguishable from burst #1’s $\varphi = 0.317 \pm 0.026$ — two FREDs of consistent asymmetry (Figure 5).

3.3. Minimum Variability Timescale

Three estimators, three labeled answers (Figure 6): the windowed framework of Bala et al. (2026) — the engine-selected canonical value — finds 30.4 ± 2.1 ms ($\Delta t = 2$ s segment, detection); the global CWT (Vianello et al. 2018) returns 215 ± 17 ms, which we note is grid-quantized (the discrete wavelet-scale ladder makes cross-burst CWT values cluster on rungs; the quoted error is grid spacing); and the Haar cross-check (Golkhou & Butler 2014) yields only an upper limit (< 1.02 s), drawn as one. In the rest frame the canonical MVT is ≈ 8 ms — a direct engine-variability timescale at $z = 2.77$.

3.4. Spectral Lag

Using the DCCF (Band 1997) with an asymmetric-Gaussian peak model (Bernardini et al. 2015) and flux-randomization Monte Carlo (Peterson et al. 1998), we scan the fit window around the correlation peak (the peak position is the deliverable; the window spread enters as a systematic): $\tau = 0.474^{+0.171}_{-0.269} \pm 0.377_{\text{win}}$ s, soft lagging hard (Norris et al. 1996; Ukwatta et al. 2010), with a 30σ correlation peak (Figure 7). The window systematic dominates: this burst’s flat-topped DCCF localizes its peak only to ± 0.4 s, and any use of this lag must carry that width. In lag-width terms, $\tau/T_{90} \approx 0.042$ — half of burst #1’s 0.085: two bursts into the campaign, the lag-width ratio is already not constant. Rest-frame lag ≈ 0.13 s.

4. TIME BINNING

Bayesian blocks (Scargle et al. 2013) through 3ML (Vianello et al. 2015) with the campaign’s $S \geq 10$ sig-

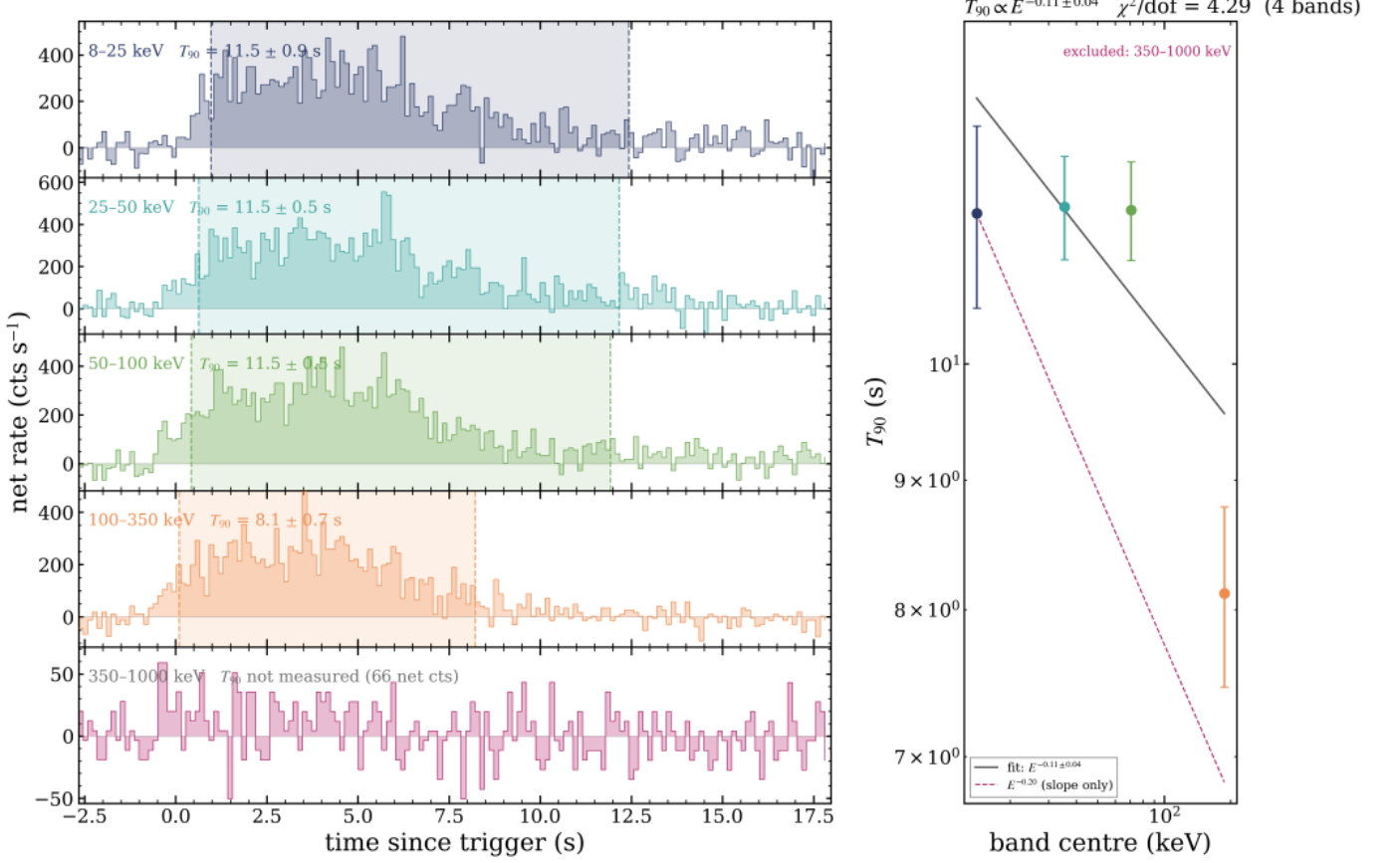
bn081222204 — step 7: energy-resolved T_{90} (detector n1); dashed = t_5 , t_{95} 

Figure 4. Per-band light curves with individual T_{90} spans (the 350–1000 keV band is excluded for insufficient counts) and the $T_{90}(E)$ relation; the population slope of Qin et al. (2013) is shown for reference only.

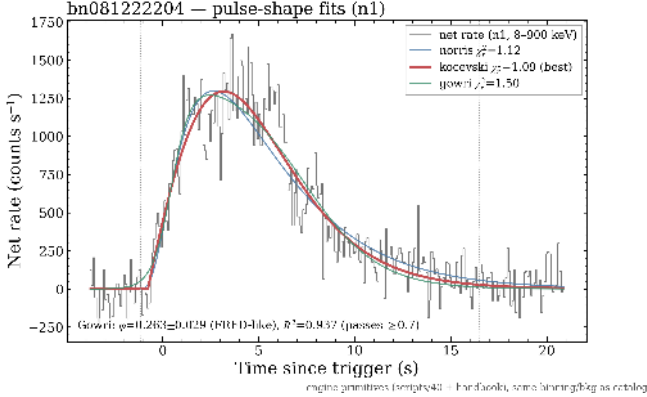


Figure 5. Pulse-shape fits to the net 8–900 keV light curve; the Kocevski profile is the χ^2_r winner.

nificance merge yield six bins spanning -0.47 to 14.57 s (Figure 8).

5. TIME-RESOLVED SPECTROSCOPY

5.1. Method and Panel Provenance

We fit the campaign’s 24 spectral models in each bin and the integrated spectrum (PGstat minimiza-

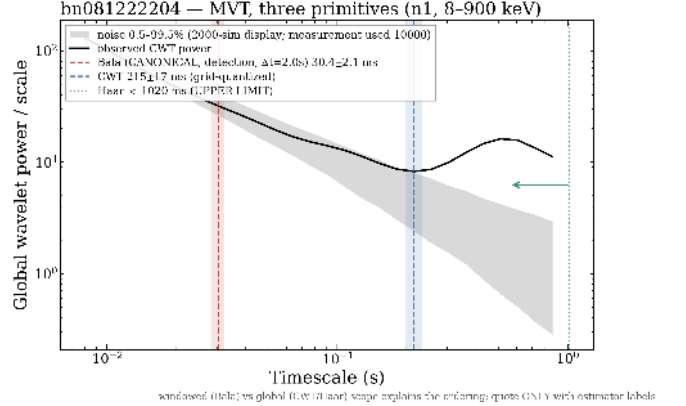


Figure 6. Global CWT power spectrum against its Poisson floor with all three MVT estimators marked; the Haar value is an upper limit and is rendered as one. Windowed-vs-global scope explains the ordering.

tion, AIC selection (Akaike 1974); $\Delta\text{AIC} < 2$ reported as ties). Every (model, bin) pair is rendered under the campaign’s no-model-dropped rule in one of three provenance tiers: live-verified (the display fit reproduces the stored solution to $|\Delta\text{AIC}| \leq 0.1$), frozen re-

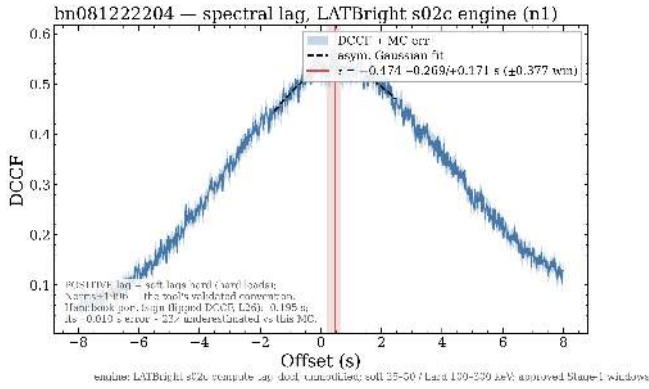


Figure 7. DCCF of the 25–50 vs. 100–300 keV light curves with the window-scanned asymmetric-Gaussian peak fit; the legend carries both the Monte-Carlo errors and the window systematic.

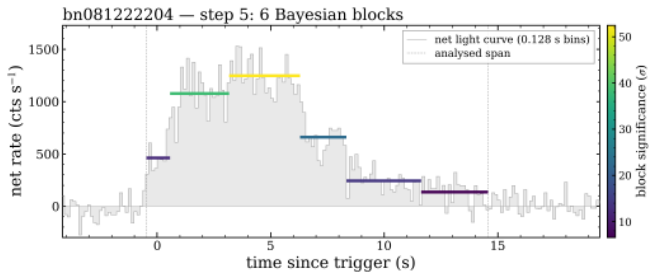


Figure 8. Bayesian-block decomposition and the six spectroscopy bins.

play (the stored solution evaluated exactly, stamped, no error band), or a labeled structural refusal (none occurred: 168/168 panels exist). One model required an engine retry in the faintest bin (a transient multi-start failure); it converged on the second attempt and its row was merged. The time-integrated winner has $\text{PGstat}/\text{dof} = 5593.9/460$.

5.2. Results

Table 1 lists winners and parameters. Figure 9 shows the integrated spectrum; Figure 12 the AIC-ordered 24-model montage; Figures 13–15 the parameter evolution of the three winner-union models; per-bin montages are in Appendix A.

The tail owns this burst. Band wins the integrated spectrum outright ($\beta = -2.24^{+0.09}_{-0.12}$; plain CPL trails by 13.3, driven by a coherent BGO 1–4 MeV run) and the hard rise (bin 0: $\alpha = +0.06^{+0.51}_{-0.37}$, $\beta = -1.69^{+0.09}_{-0.14}$) and mid-burst (bin 2; the hard-rise spectrum is shown in Figure 10). CPL suffices in the softer bins. Unlike burst #1, no composite wins the integrated spectrum: with a gentler E_p decay (152 $\rightarrow$ 104 keV; Figure 14), time integration does not manufacture spurious components here.

Synchrotron-compatible continuum. Across the body of the burst the winning-model α spans -0.70 to -0.96 — below the $-2/3$ line of death, the opposite of burst #1’s $\alpha \simeq -0.3$. With the b_0 effective-area rail as a standing caveat, this burst’s respect of the synchrotron limit is testable now; burst #1’s violation requires a calibration-decoupling refit before it can be claimed (see §6).

A physical-temperature thermal candidate, honestly weighed. In the decay bin 4 ([8.3, 11.6] s), CPL+BB tops all 24 models ($\Delta\text{AIC} = 0.8$ over second) with $kT = 16.5 \pm 3.7$ keV — Wien peak $\simeq 65$ keV, well inside the fitted band, so it passes the campaign’s edge-artifact gate (which burst #1’s spurious low- kT winners fail). An independent earlier analysis of the same phase found $kT = 21.4$ keV, a consistency worth noting. But the residual evidence is thin: no localized 40–100 keV residual mode distinguishes CPL+BB from plain CPL (Figure 11, where the dotted decomposition shows the BB contributing a shallow shoulder under a continuum three times higher), the likelihood gain sits in continuum re-shaping (the cutoff runs unconstrained), the BB normalization is $\sim 1\sigma$ from zero, and kT shows no continuity in neighboring bins (Figure 15). We record it as a *tracked candidate* ($N = 1$), not a detection.

6. DISCUSSION

Two bursts, opposite spectra. The campaign’s first two FREDs of nearly equal asymmetry disagree on both headline spectral questions: burst #1 sits at $\alpha \simeq -0.3$ (violating the synchrotron line of death in every bin) with zero thermal evidence and an integrated composite that is evolution in disguise; GRB 081222 respects the limit everywhere, hosts one physical-temperature thermal candidate, and its integrated spectrum is an honest Band. A two-burst “population” proves nothing — but it already demonstrates that single-pulse morphology does not pin down the emission physics, and it sets up the campaign’s per-burst α census with a calibration caveat attached: in both bursts one detector’s effective-area factor rails at its bound, and the burst-#1 violation claim is gated on a decoupling refit before it enters any census.

The thermal question, done both ways. Burst #1 shows how thermal components die at the band edge (its bin-2 “winner” has $3.92 kT < 8$ keV); GRB 081222 shows the opposite failure mode: a candidate that passes the edge gate and still cannot claim detection because the residuals do not demand it. The pair calibrates

Table 1. AIC-selected best model per bin.

Bin	Interval (s)	Model	AIC	$\Delta\text{AIC}_{2\text{nd}}$	Parameters
T_{INT}	$[-0.47, 14.57]$	Band	5607.9	1.1	$\alpha = -0.81^{+0.05}_{-0.05}$, $E_p = 150^{+10}_{-9}$, $\beta = -2.24^{+0.09}_{-0.12}$
0	$[-0.47, 0.58]$	Band	1800.6	0.7	$\alpha = +0.06^{+0.51}_{-0.37}$, $E_p = 152^{+46}_{-32}$, $\beta = -1.69^{+0.09}_{-0.14}$
1	$[0.58, 3.17]$	CPL	3222.8	1.1	$\Gamma = -0.96^{+0.05}_{-0.05}$, $E_c = 259^{+38}_{-31}$
2	$[3.17, 6.30]$	Band	3446.3	1.0	$\alpha = -0.70^{+0.06}_{-0.06}$, $E_p = 140^{+9}_{-9}$, $\beta = -2.43^{+0.13}_{-0.18}$
3	$[6.30, 8.34]$	CPL	2824.0	1.9	$\Gamma = -0.80^{+0.11}_{-0.11}$, $E_c = 105^{+20}_{-15}$
4	$[8.34, 11.63]$	CPL+BB ^a	3302.1	0.8	$\Gamma = -1.62 \pm 0.11$, $E_c = 3227 \pm 4500$; $kT = 16.5 \pm 3.7$, $K_{\text{BB}} = (6.4 \pm 5.5) \times 10^{-5}$
5	$[11.63, 14.57]$	CPL	3196.6	1.8	$\Gamma = -0.90^{+0.35}_{-0.31}$, $E_c = 104^{+93}_{-38}$

^aPhysical-temperature candidate (passes the Wien-peak edge gate) with thin residual evidence; reported as tracked, not detected. MINOS failed for this fit; errors are covariance-based.

NOTE—Energies in keV; all margins to second place are ties or near-ties ($\Delta\text{AIC} \leq 2$). Full 24-model tables are machine-readable products. All values provisional pending campaign freeze.

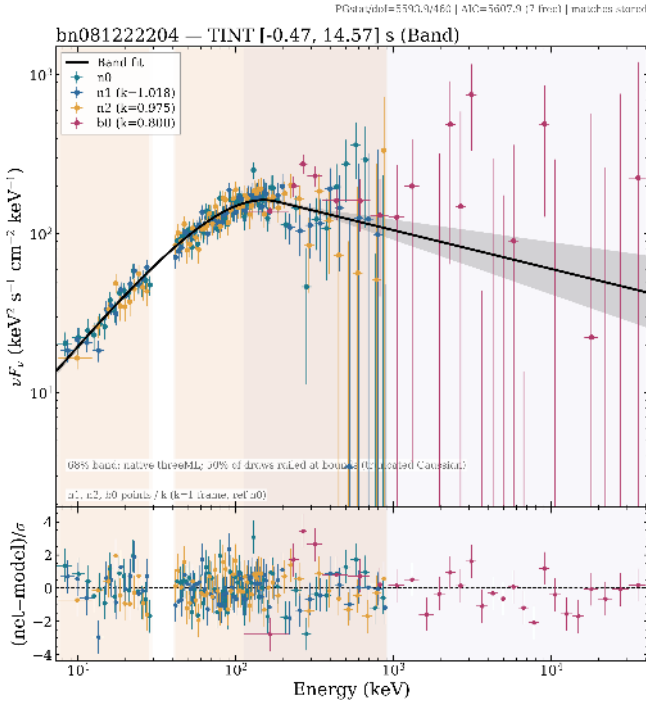


Figure 9. Time-integrated νF_ν spectrum with the winning Band fit (strict-XSPEC unfolding at **rebin 5 5**; 68% band from native 3ML propagation with validity guards; count-space residuals below).

the campaign’s census instrument in both directions — specificity and restraint.

Timing quantities carry their estimators. The MVT spans 30 ms–1 s on the same data depending on the question asked; the lag’s window systematic exceeds its statistical error. Rest-frame values ($T_{90} \simeq 3.0$ s, MVT $\simeq 8$ ms, $\tau \simeq 0.13$ s) enter the campaign’s lag-variability program (Shenoy et al. 2013; Göktaş et al.

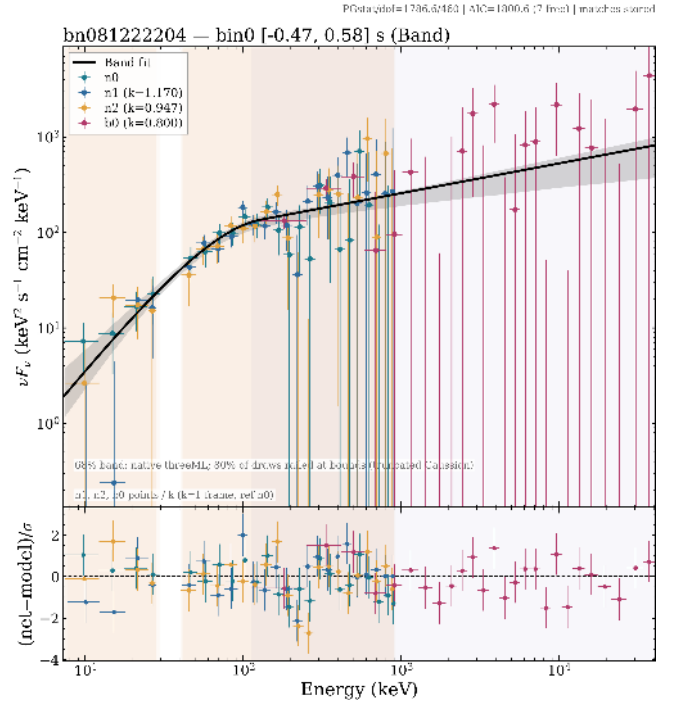


Figure 10. The hard rise (bin 0): Band fit with $\alpha = +0.06$ and $\beta = -1.69$ — the tail present from the first second.

2025) with those labels attached — and the already-diverging lag-width ratios of the first two bursts (0.085 vs. 0.042) preview what a single-pulse-selected sample can say that mixed-morphology samples cannot.

7. ANALYSIS PROVENANCE

Every figure passed a fresh-context verification gate before inclusion (producer-verifier separation, sha256-bound verdicts); every number binds to a machine-readable sidecar; live display fits are hard-guarded to the stored solutions; all 168 model panels exist under the

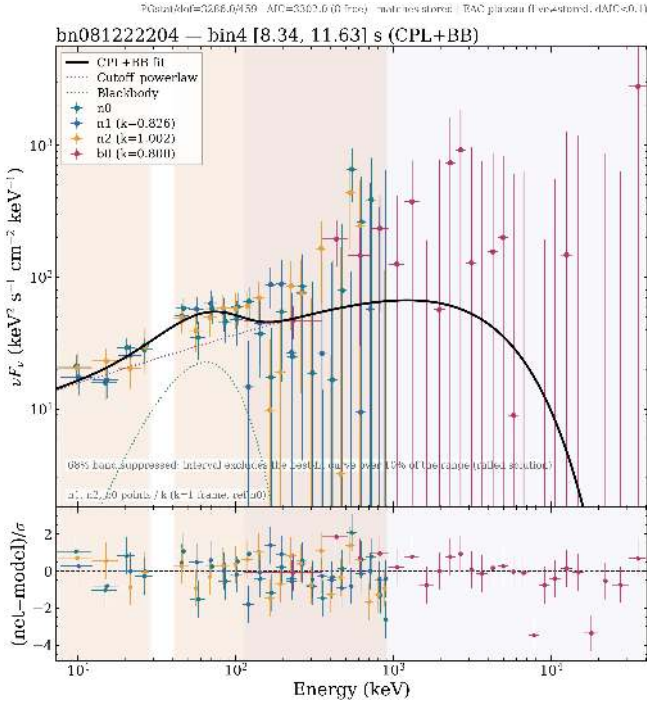


Figure 11. The bin-4 CPL+BB candidate with XSPEC-style dotted component decomposition: the $kT = 16.5$ keV blackbody contributes a shallow ~ 65 keV shoulder beneath a continuum three times higher — the graphic form of the “likelihood-thin” verdict. The 68% band is suppressed here because the bound-railed solution’s interval fails the curve-containment guard (stated on-figure).

no-model-dropped rule; the delivery pipeline is hook-enforced (un-gated figures are mechanically blocked). The gate ledger and per-bin reviewer notes are archived

with the campaign products; a companion paper describes the agentic pipeline.

8. SUMMARY

1. $T_{90} = 11.33 \pm 0.32$ s (windowed; window captures the burst; rest frame 3.01 s); $T_{90}(E)$ follows a single power law poorly ($\chi^2/\text{dof} = 4.3$); $\varphi = 0.263 \pm 0.029$ (FRED).
2. MVT = 30.4 ± 2.1 ms (windowed canonical; ≈ 8 ms rest frame) vs. 215 ± 17 ms (global CWT, grid-quantized) and < 1.02 s (Haar limit) — estimator labels are part of the measurement.
3. Lag = $0.474^{+0.171}_{-0.269} \pm 0.377_{\text{win}}$ s: the window systematic dominates; $\tau/T_{90} = 0.042$, half of burst #1’s ratio.
4. $\alpha = -0.70$ to -0.96 : synchrotron-compatible throughout — the opposite of burst #1; Band wins the integrated spectrum outright with $\beta = -2.24$.
5. A physical-temperature thermal candidate ($kT = 16.5 \pm 3.7$ keV, decay phase) passes the edge gate but fails the residual test: tracked, not detected.
6. All 168 (model, bin) panels exist under the no-model-dropped rule; every figure and number carries verified provenance.

Fermi/GBM public data; analysis with 3ML (Vianello et al. 2015). Produced by the single-pulse campaign’s staged, gate-reviewed agentic pipeline; all numbers provisional pending the campaign freeze.

APPENDIX

A. PER-BIN MODEL MONTAGES

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Figure 12. All 24 models fit to the integrated spectrum, AIC-ordered (winner framed). Every cell is a real panel under the no-model-dropped rule; frozen replays are stamped.

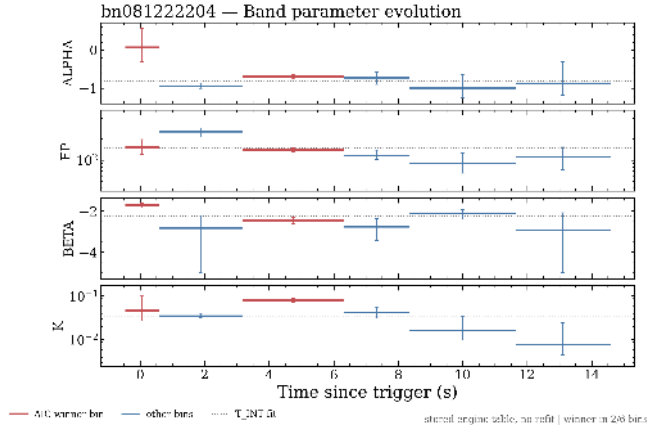


Figure 13. Band parameter evolution (red = winner bins).

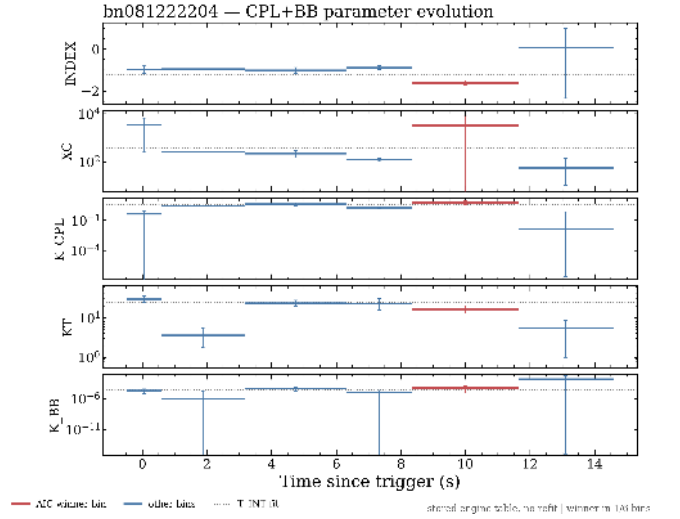


Figure 15. CPL+BB parameter evolution; kT shows no continuity across bins — part of the candidate verdict.

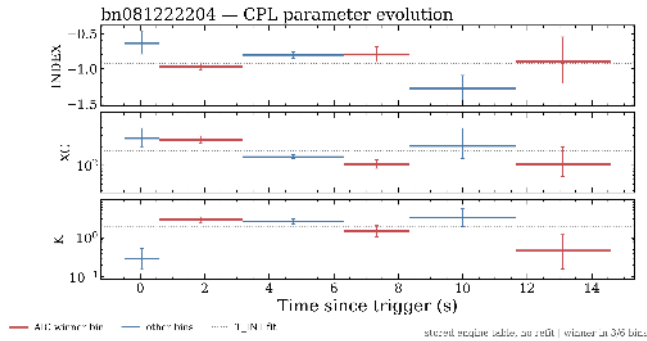


Figure 14. CPL parameter evolution: E_c decays $259 \rightarrow 104$ keV while Γ stays near -0.9 .

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bn081222204 — bin0 [-0.47, 0.58] s — 24 models, AIC-ordered — winner BAND

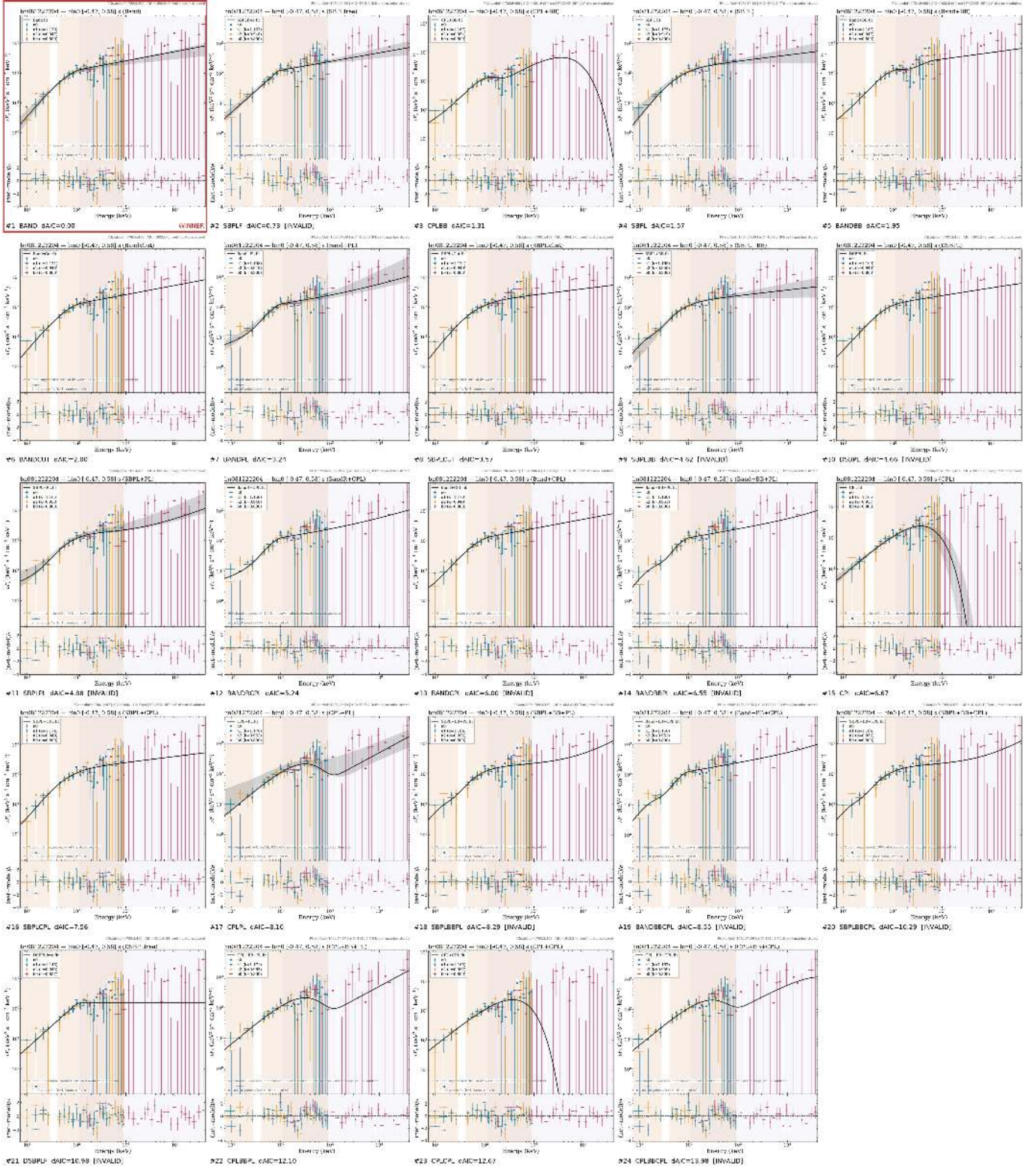


Figure 16. Bin 0 montage.

bn01222204 — bin1 [0.58, 3.17] s — 24 models, AIC-ordered — winner CPL

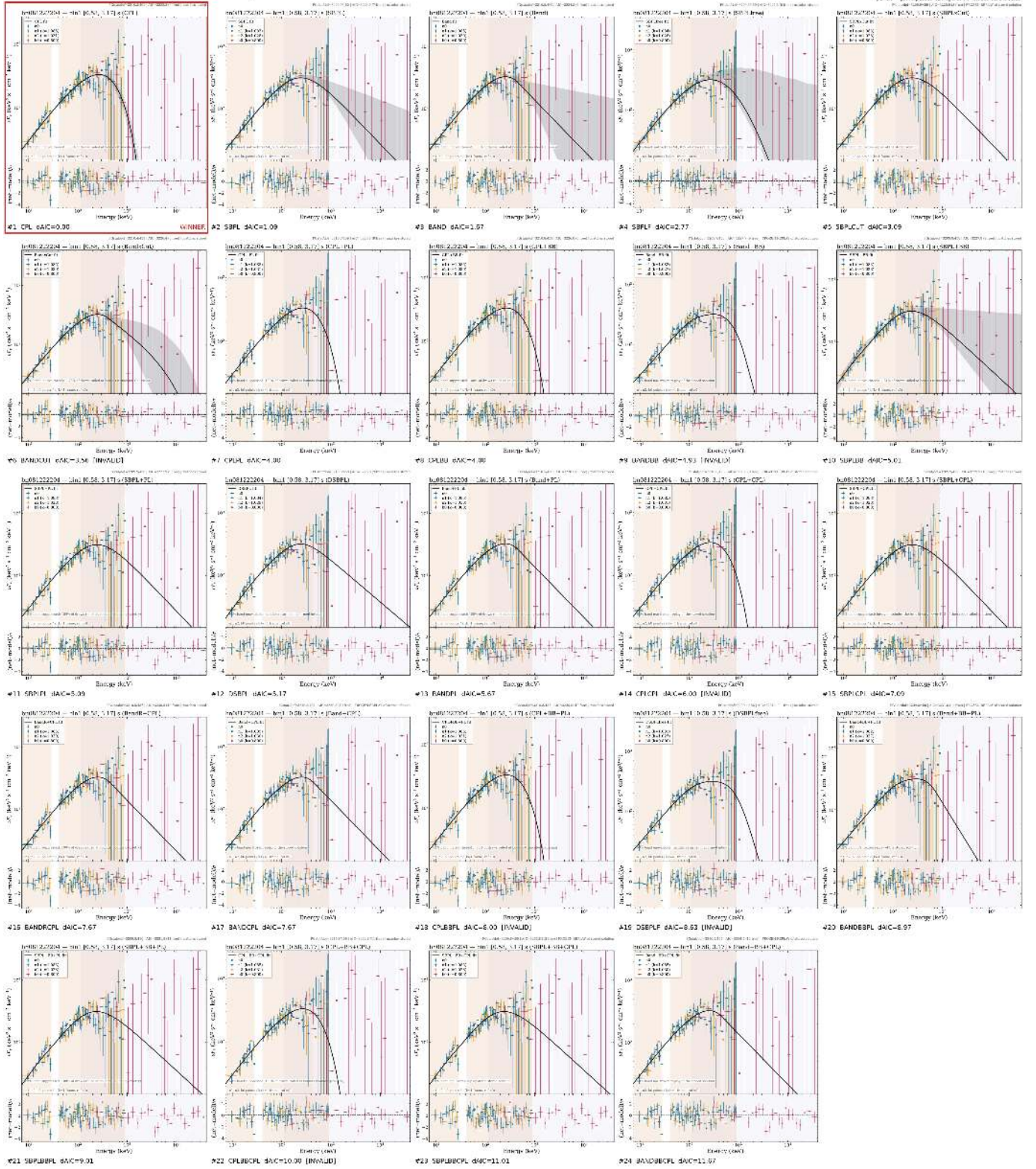


Figure 17. Bin 1 montage.

bn081222204 — bin2 [3.17, 6.30] s — 24 models, AIC-ordered — winner BAND

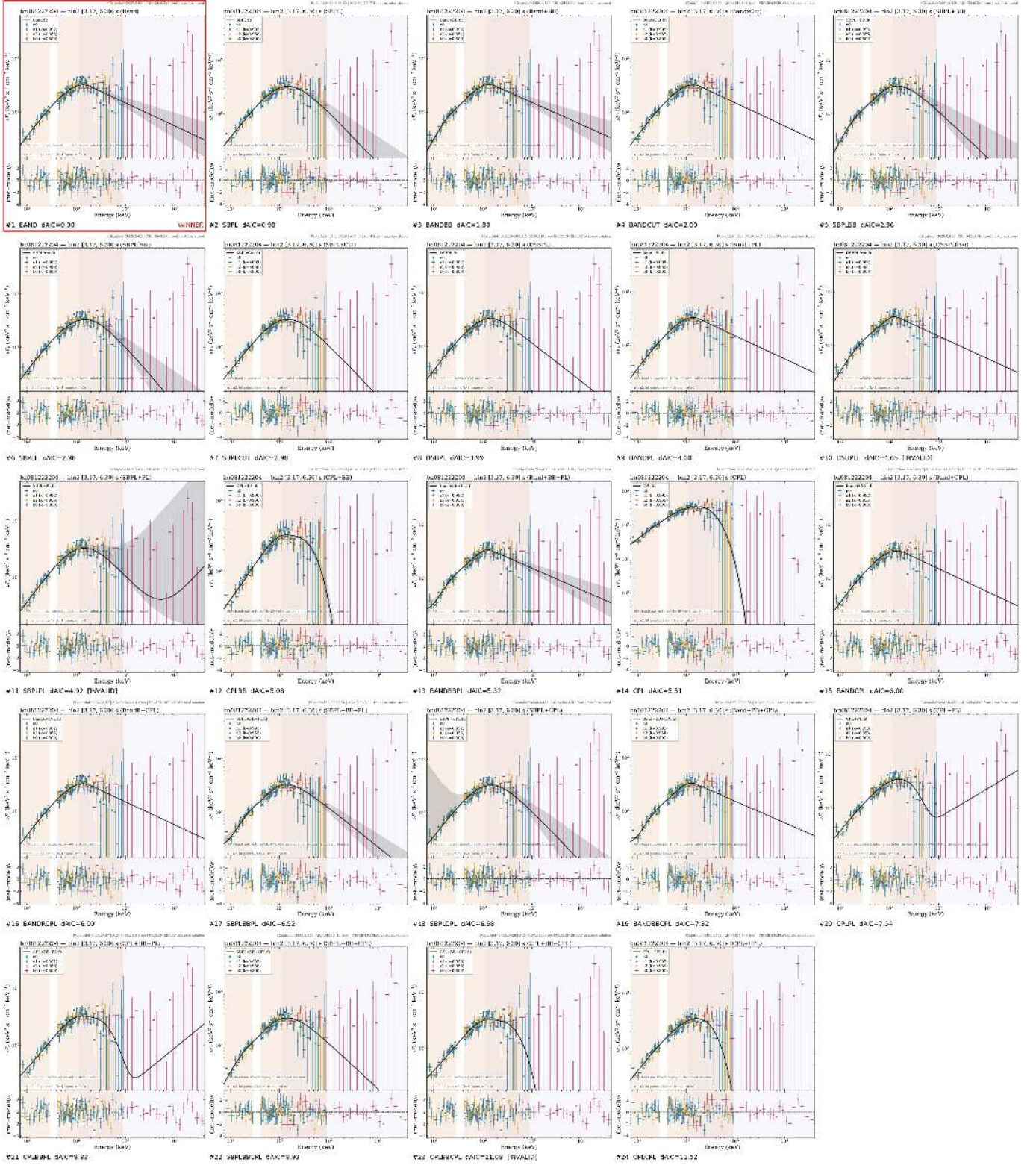


Figure 18. Bin 2 montage.

bn081222204 — bin3 [6.30, 8.34] s — 24 models, AIC-ordered — winner CPL

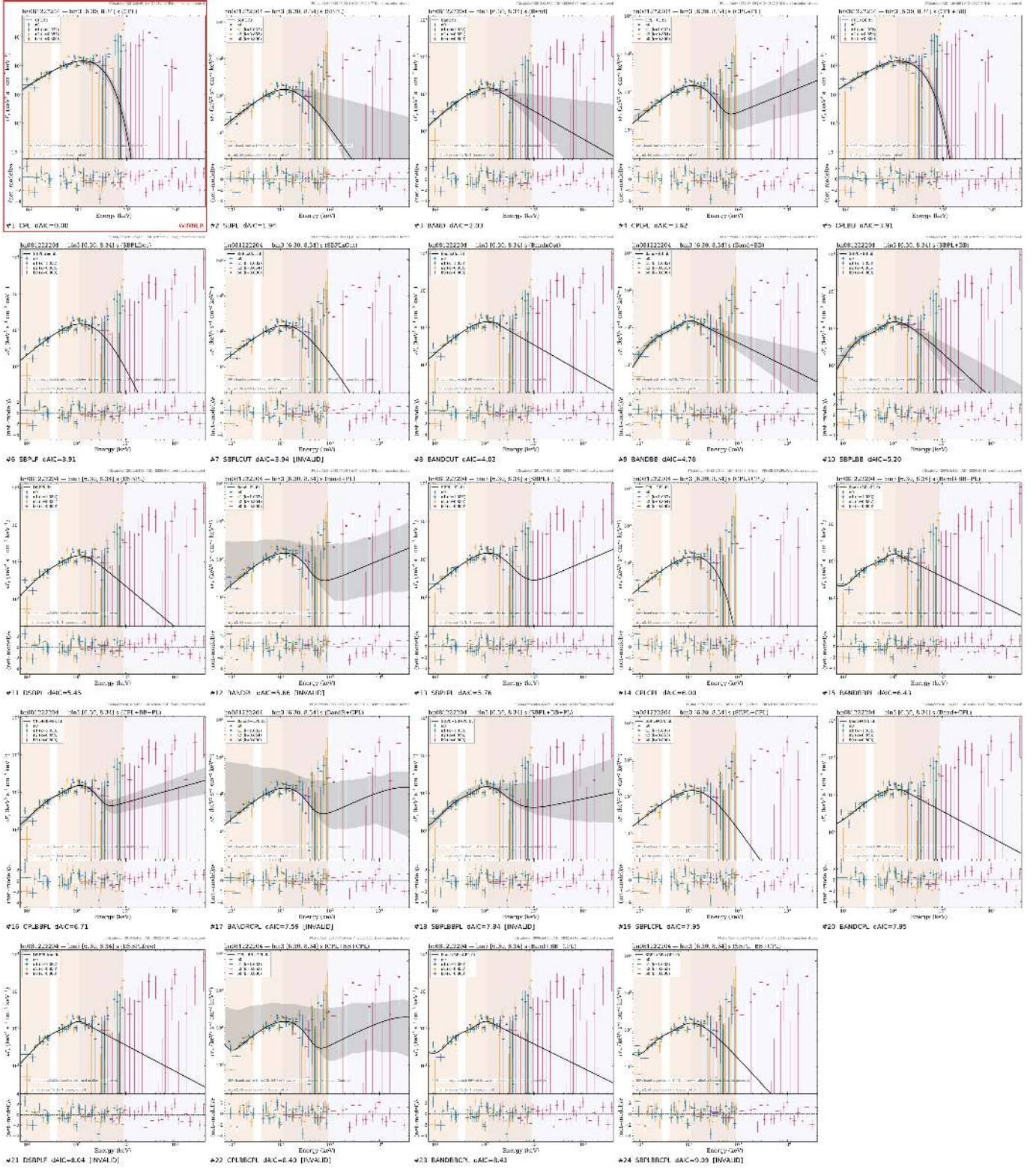


Figure 19. Bin 3 montage.

bn081222204 — bin4 [8.34, 11.63] s — 24 models, AIC-ordered — winner CPLBB

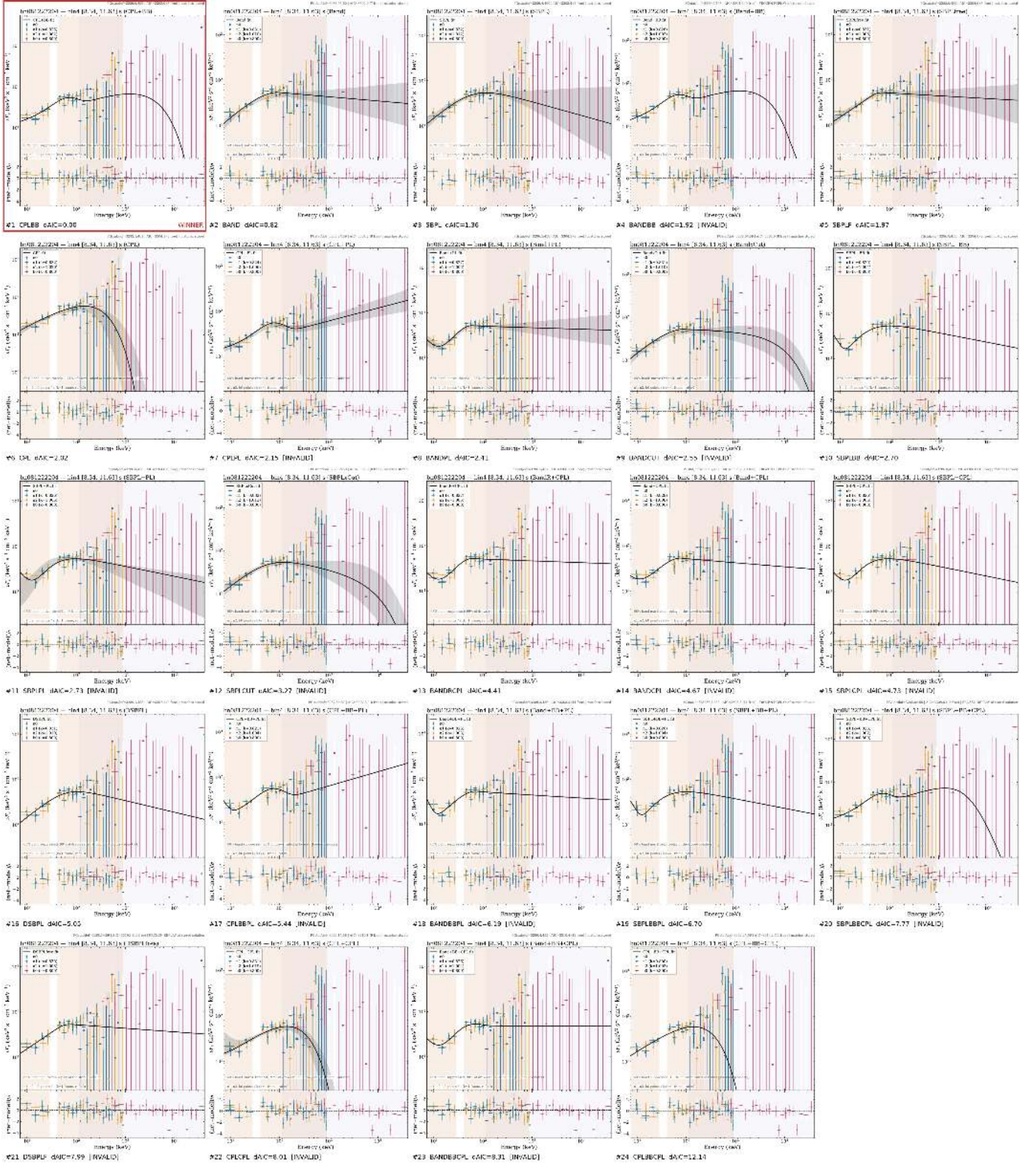


Figure 20. Bin 4 montage.

bn081222204 — bin5 [11.63, 14.57] s — 24 models, AIC-ordered — winner CPL

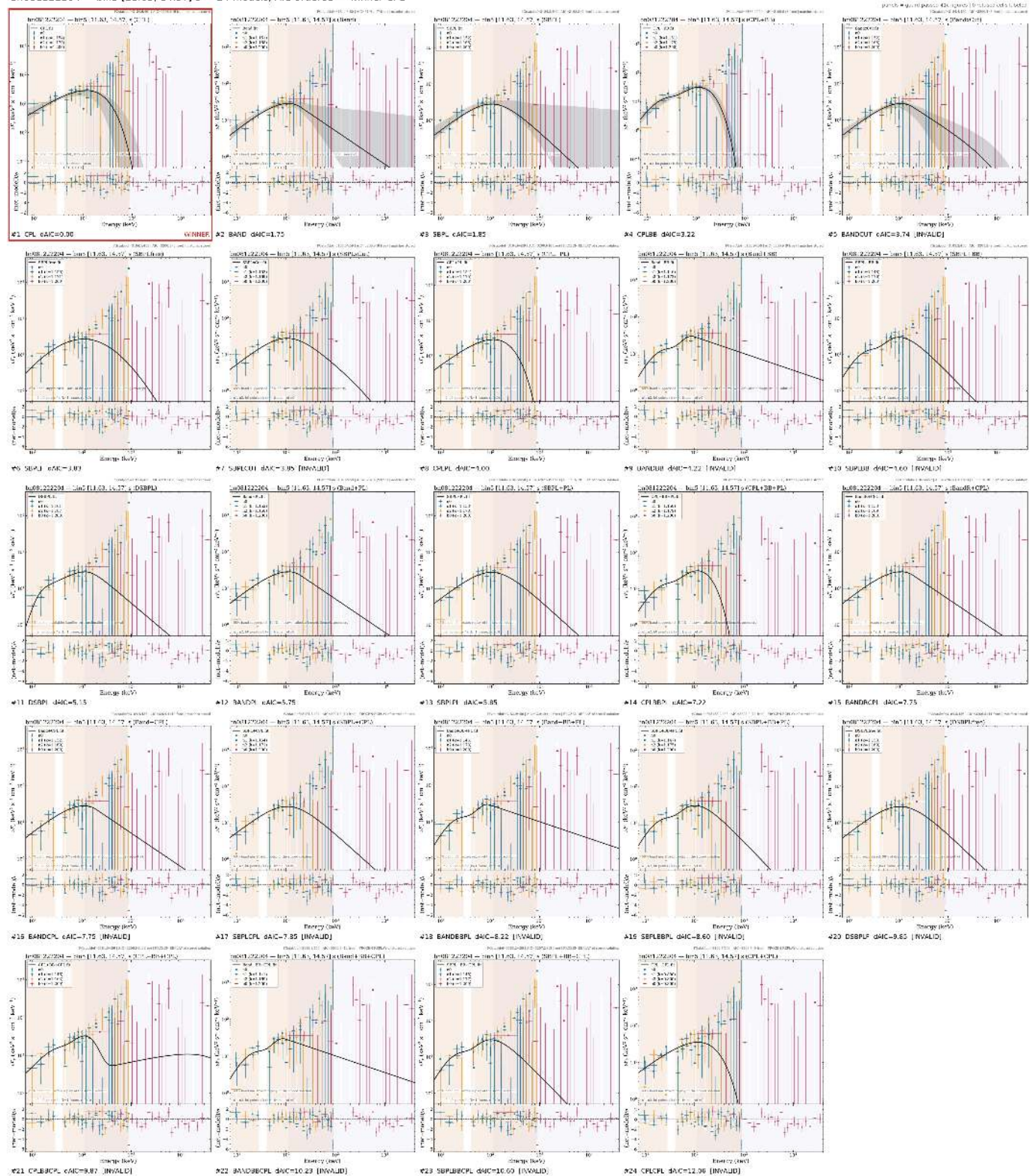


Figure 21. Bin 5 montage.